SaaS Customer Churn Analysis Project

1. Project Overview

This project analyzes customer churn in a B2C SaaS subscription business using customer, usage, payment, and feedback data.

The objective is to identify churn patterns, understand factors influencing customer retention, and generate actionable insights that help improve customer lifetime value and revenue stability.

The analysis evaluates customer engagement, subscription plans, payment behavior, and feedback scores to detect churn trends and high-risk customer segments.

2. Business Problem

Subscription-based businesses depend heavily on customer retention for stable revenue. High churn rates can significantly reduce profitability and long-term growth.

Key business challenges include:

• Identifying which customers are most likely to churn

• Understanding which subscription plans have the highest churn

• Determining whether low product engagement leads to churn

• Analyzing the relationship between customer feedback and churn

• Estimating customer lifetime value (LTV) to guide retention strategies

The goal is to use data-driven insights to reduce churn and increase customer retention.

3. Data Source

The dataset used in this project is a synthetic SaaS subscription dataset designed to simulate real-world customer behavior.

Tables used in the analysis:

customers.csv

- customer_id
- signup_date
- country
- subscription_plan
- monthly_price
- status (Active / Churned)

usage_data.csv

- usage_id
- customer_id
- usage_month
- login_count
- feature_usage

payments.csv

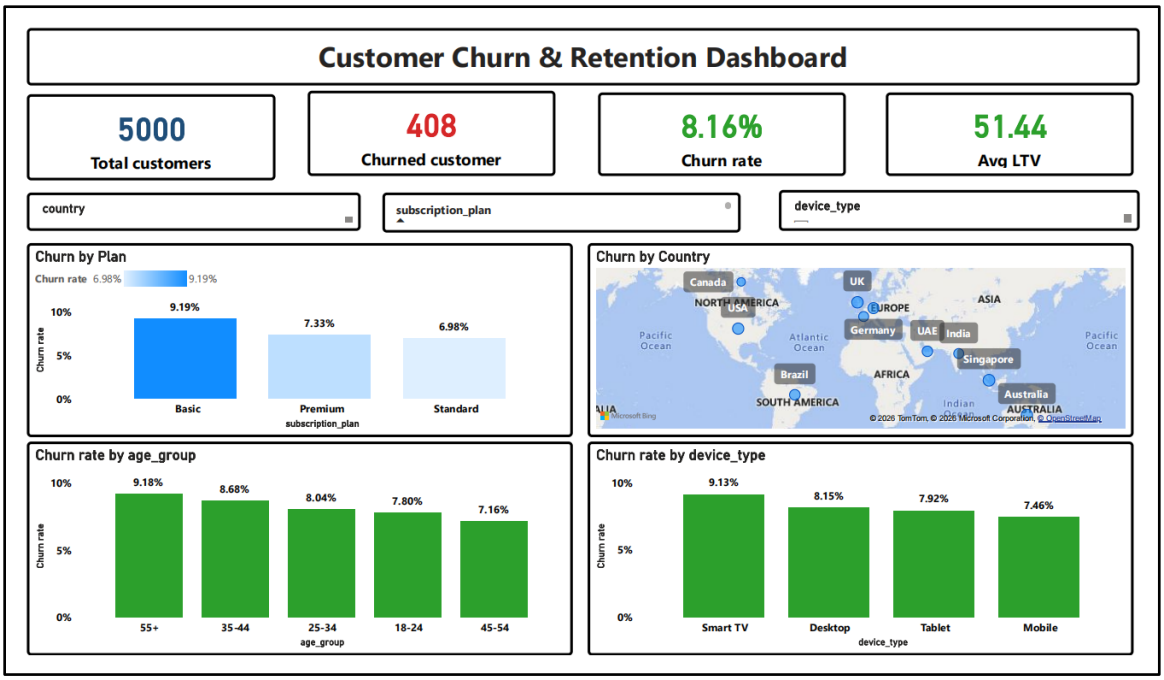
- payment_id
- customer_id
- payment_date
- amount
- payment_status

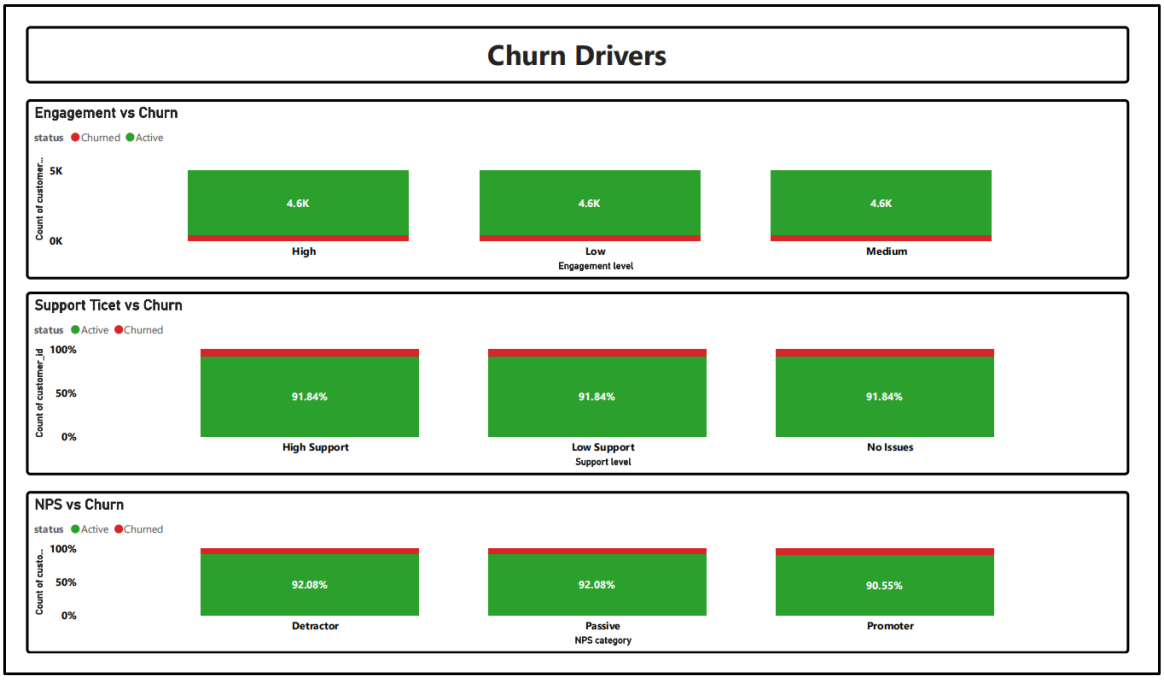
feedback.csv

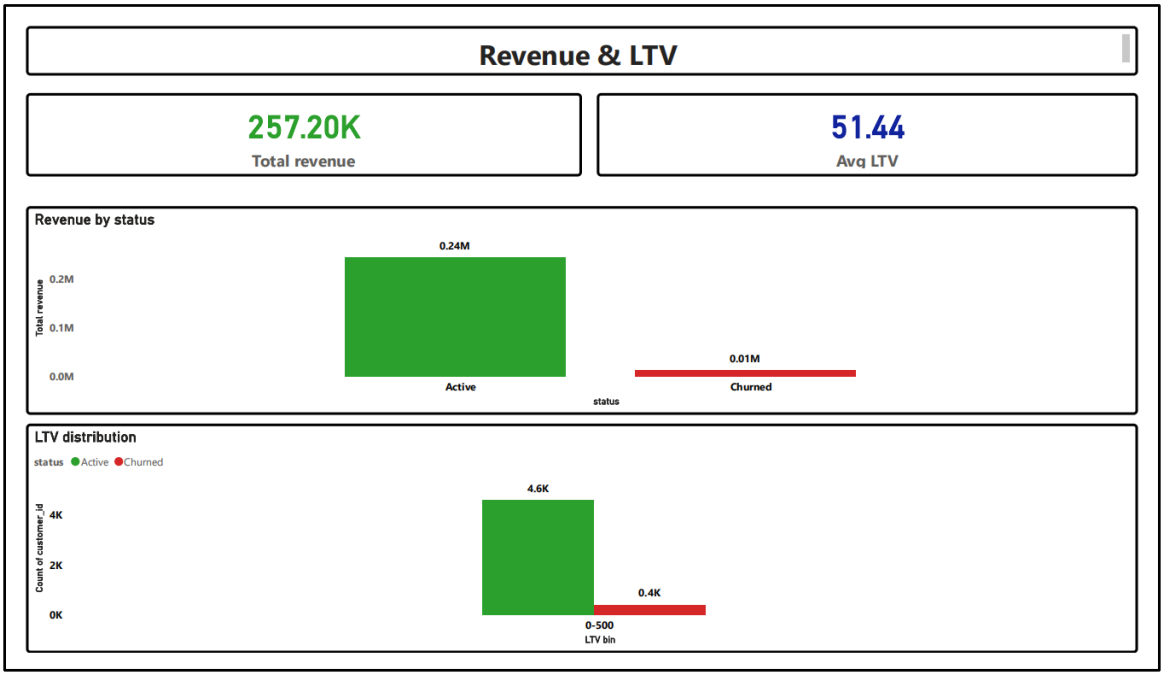
- feedback_id
- customer_id
- feedback_score
- feedback_date

4. Tools and Technologies

- MySQL – Data storage, SQL analysis, joins, churn calculations
- Power BI – Data visualization and dashboard development
- Excel / CSV – Data inspection and preparation







5. Key Metrics Tracked

Customer Churn Rate

Churn Rate = (Churned Customers / Total Customers) × 100

Monthly Recurring Revenue (MRR)

Total predictable monthly revenue from subscriptions.

Annual Recurring Revenue (ARR)

ARR = MRR × 12

Average Revenue Per User (ARPU)

ARPU = Total Revenue / Total Customers

Customer Lifetime Value (LTV)

LTV = ARPU ÷ Churn Rate

Customer Engagement Metrics

- Login frequency
- Feature usage frequency
- Product engagement levels

6. Key Insights

- Higher churn was observed in lower subscription plans.
- Customers with lower login frequency were more likely to churn.
- Customers with low feedback scores had higher churn probability.
- Certain geographic regions demonstrated higher churn rates.
- Premium plan churn had a higher revenue impact even with fewer customers.

7. Business Recommendations

- Improve customer onboarding to increase early engagement.
- Target low-usage customers with retention campaigns.

- Monitor feedback scores and proactively support dissatisfied customers.
- Offer loyalty incentives or discounts for at-risk customers.
- Encourage upgrades to higher-value plans through targeted offers.

8. Project Outcome

The churn analysis provided insights that help businesses identify churn drivers, detect high-risk customers, and design better retention strategies. The analysis supports improved revenue forecasting and customer lifecycle management.

9. Skills Demonstrated

- SQL Data Analysis
- Customer Churn Modeling
- Business Intelligence & KPI Analysis
- Data Visualization with Power BI
- Data-driven Business Decision Making